

TASK 01

Cyber Security Tasks & Instructions — CodeAlpha

SUBMITTED BY: Muhammad Omer Ibraim

Task No. 1

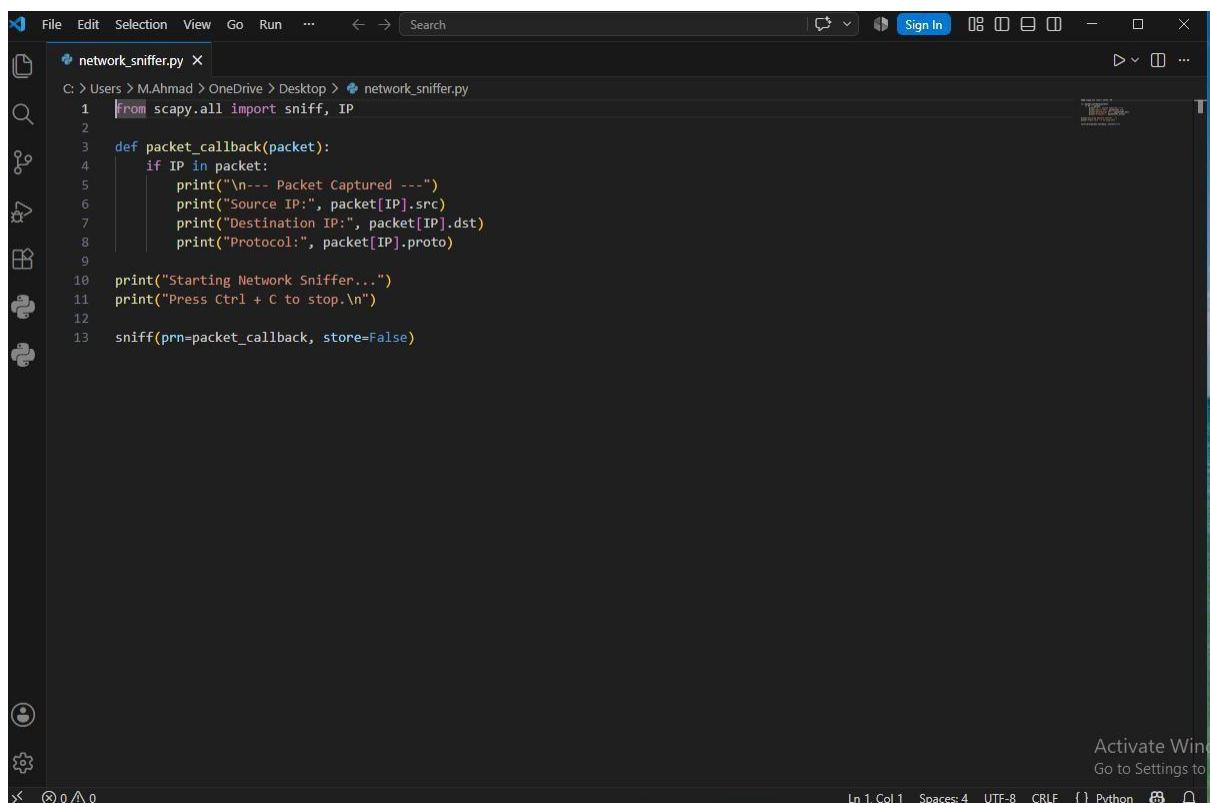
Basic Network Sniffer

1. Build a Python program to capture network traffic packets.
2. Analyze captured packets to understand their structure and content.
3. Learn how data flows through the network and the basics of protocols.
4. Use libraries like `scapy` or `socket` for packet capturing
5. Display useful information such as source/destination IPs, protocols and payloads..

Basic Network Sniffer Submission Steps:

Step 1: Python Code

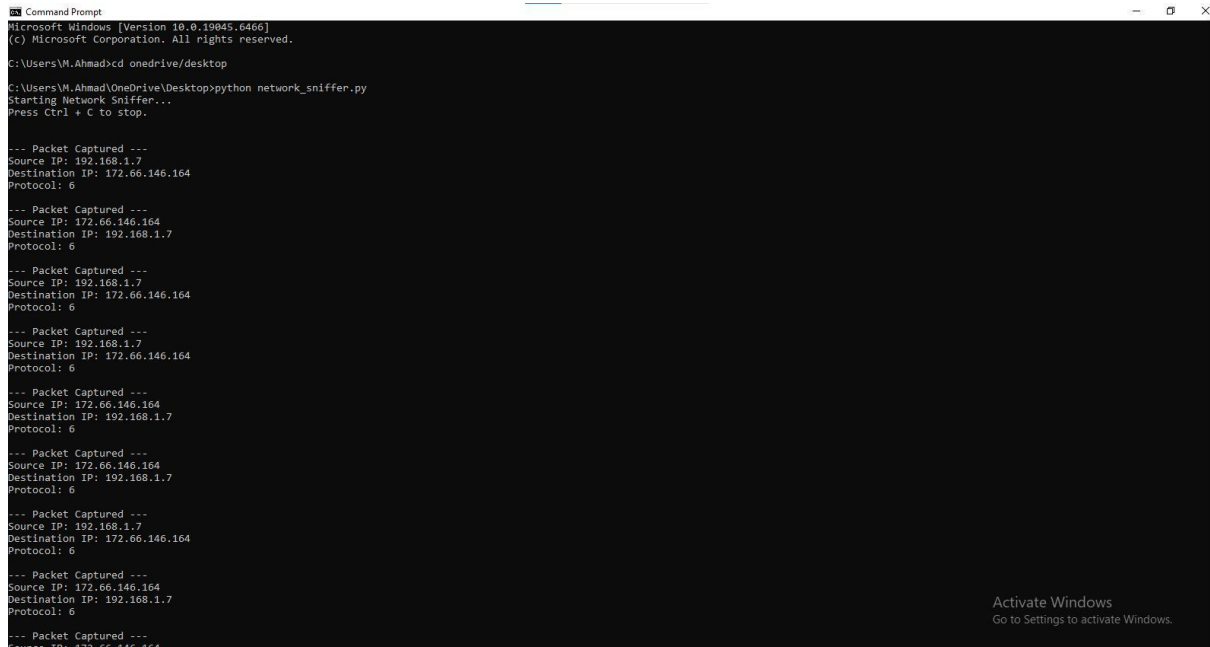
Attach a screenshot of your network_sniffer.py code.



```
1  from scapy.all import sniff, IP
2
3  def packet_callback(packet):
4      if IP in packet:
5          print("\n--- Packet Captured ---")
6          print("Source IP:", packet[IP].src)
7          print("Destination IP:", packet[IP].dst)
8          print("Protocol:", packet[IP].proto)
9
10     print("Starting Network Sniffer...")
11     print("Press Ctrl + C to stop.\n")
12
13     sniff(prn=packet_callback, store=False)
```

Step 2: Program Output

Attach a screenshot showing the network packets captured by the program..



```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\W.Ahmad>cd onedrive/desktop
C:\Users\W.Ahmad\OneDrive\Desktop>python network_sniffer.py
Starting Network Sniffer...
Press Ctrl + C to stop.

--- Packet Captured ---
Source IP: 192.168.1.7
Destination IP: 172.66.146.164
Protocol: 6

--- Packet Captured ---
Source IP: 172.66.146.164
Destination IP: 192.168.1.7
Protocol: 6

--- Packet Captured ---
Source IP: 192.168.1.7
Destination IP: 172.66.146.164
Protocol: 6

--- Packet Captured ---
Source IP: 192.168.1.7
Destination IP: 172.66.146.164
Protocol: 6

--- Packet Captured ---
Source IP: 172.66.146.164
Destination IP: 192.168.1.7
Protocol: 6

--- Packet Captured ---
Source IP: 172.66.146.164
Destination IP: 192.168.1.7
Protocol: 6

--- Packet Captured ---
Source IP: 192.168.1.7
Destination IP: 172.66.146.164
Protocol: 6

--- Packet Captured ---
Source IP: 172.66.146.164
Destination IP: 192.168.1.7
Protocol: 6

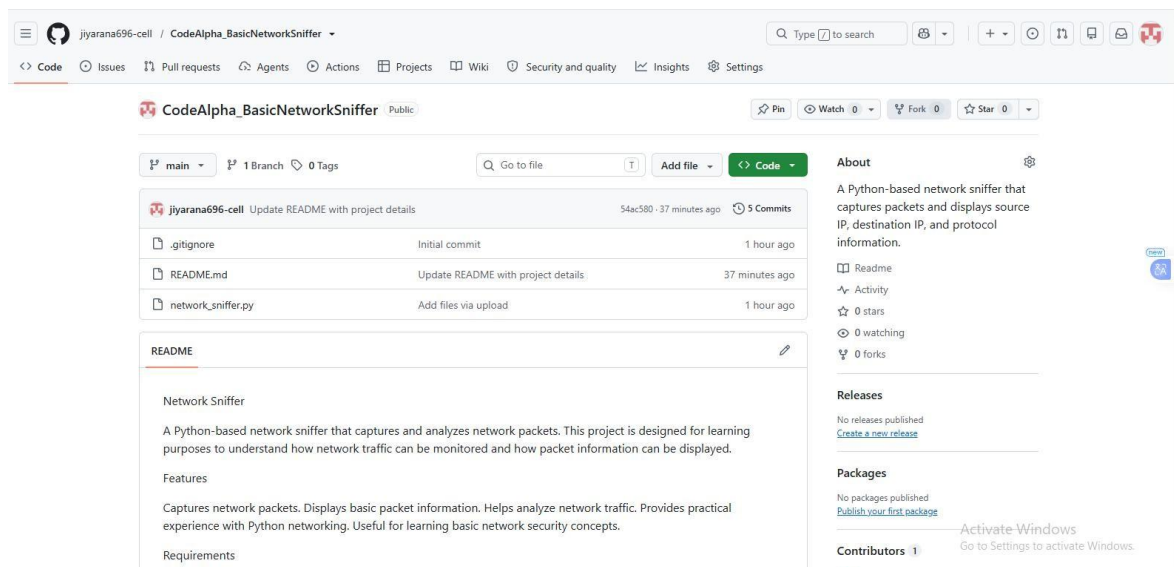
--- Packet Captured ---
Source IP: 172.66.146.164
Destination IP: 192.168.1.7
Protocol: 6

--- Packet Captured ---
Source IP: 172.66.146.164
Destination IP: 192.168.1.7
Protocol: 6
```

Step 3 : GitHub Repository

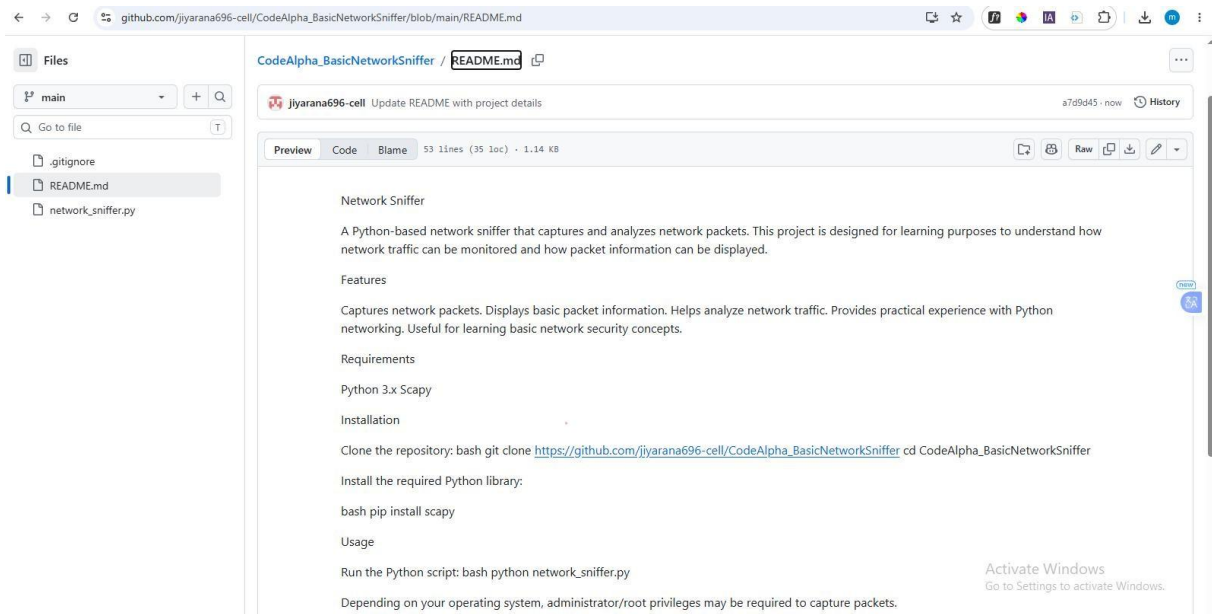
Attach a screenshot of your GitHub repository showing:

- network_sniffer.py
- README.md
- .gitignore



Step 4 : README.md

Attach a screenshot of your completed README.md:



Step 5 : GitHub Repository Link

Submit your GitHub repository link:

https://github.com/jiyarana696-cell/CodeAlpha_BasicNetworkSniffer